CSE400 Lecture 18 Scribe

L18.pdf
PDF

EXAMPLE LECTURE SCRIBE

Definitions and Notations

a) Definition:
Marginal PMF is obtained by summing the joint PMF over all possible values of the other variable.

b) Notation:

p
X
Y
(
x
,
y
)
p
XY

(x,y): Joint PMF

p
X
(
x
)
=
$\sum$
y
p
X
Y
(
x
,
y
)
p
X

(x)=$\sum$
y

p
XY

(x,y)

p
Y
(
y
)
=
∑
x
p
X
Y
(
x
,
y
)
p
Y

(y)=∑
x

p
XY

(x,y)

Assumptions and Preliminaries

X and Y are discrete random variables with a valid joint PMF such that total probability sums to 1.

Main Result

a) Proof:
Starting from the joint PMF
p
X
Y
(

x
,
y
)
p
XY

(x,y), summing over all values of Y gives:

p
X
(
x
)
=
∑
y
p
X
Y
(
x
,
y
)
p
X

(x)=∑
y

p
XY

(x,y).
Similarly, summing over X gives:

p
Y
(
y
)
=
∑
x
p
X

Y
(
x
,
y
)
p
Y

(y)=∑
x

p
XY

(x,y).

b) Conclusion:
Marginal distributions are obtained by eliminating the other variable through summation.

Working Examples

Example:
Given joint PMF table:

p
X
Y
(
0
,
0
)
=
1
/
8
,

p
X
Y
(
0
,
1
)

=
1
/
4
,

p
X
Y
(
1
,
0
)
=
3
/
8
,

p
X
Y
(
1
,
1
)
=
1
/
4
p
XY

(0,0)=1/8,p
XY

(0,1)=1/4,p
XY

(1,0)=3/8,p
XY

(1,1)=1/4

Steps:

p
X
(
0
)
=
1
/
8
+
1
/
4
=
3
/
8
p
X

(0)=1/8+1/4=3/8

p
X
(
1
)
=
3
/
8
+
1
/
4
=
5
/
8
p
X

(1)=3/8+1/4=5/8

p
Y
(
0

)
=
1
/
8
+
3
/
8
=
1
/
2
p
Y

(0)=1/8+3/8=1/2

p
Y
(
1
)
=
1
/
4
+
1
/
4
=
1
/
2
p
Y

(1)=1/4+1/4=1/2

Final Result:

p
X
=
{
3
/
8

,
5
/
8
}
,
p
Y
=
{
1
/
2
,
1
/
2
}
p
X

={3/8,5/8},p
Y

={1/2,1/2}

Task:
Using the same structure as the example above, generate a lecture scribe intended strictly as exam-oriented reference material for:

CSE400 – Fundamentals of Probability in Computing
Lecture 18: Marginal PMF, Correlation, Covariance, Joint Gaussian Random Variables

Use ONLY the provided Lecture 18 PDF as context.

Follow this structure exactly:

Definitions and Notations
a) Definition
b) Notation

Assumptions and Preliminaries

Main Result
a) Proof
b) Conclusion

Working Examples

The scribe must faithfully cover all material presented in the Lecture 18 PDF, including marginal PMF, correlation, covariance, Pearson's coefficient, and jointly Gaussian random variables.

Include definitions, notation, assumptions, results, proofs or derivations, conclusions, and worked examples ONLY if they explicitly appear in the Lecture PDF.

Do NOT summarize, explain intuitively, simplify, reorganize content beyond the stated structure, or introduce any material, examples, or interpretations not present in the Lecture PDF.

The output must be a faithful reconstruction of the lecture and suitable for closed-notes exam revision.

give the lecture scribe according to the prompt given above regarding the lecture pdf i have attatched

Here is the **lecture scribe strictly following your required structure and using only the provided PDF**

CSE400 – Lecture 18 Scribe

Definitions and Notations

a) Definition

Marginal PMF:
The marginal PMF of one variable is obtained by summing the joint PMF over all possible values of the other variable.

Correlation (Product Moment):
A basic measure of joint behavior defined as the expected value of the product of two variables:

$$R_{XY} = E[XY]$$

It represents the average value of the product XY.

Covariance:
Covariance measures how two variables vary together after removing their means:

$$\mathrm{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

It captures whether deviations from means occur in the same or opposite direction.

Pearson's Correlation Coefficient:
A normalized measure of covariance:

$$\rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y}$$

It measures the strength and direction of linear relationship.

Jointly Gaussian Random Variables:
A pair (X,Y) is jointly Gaussian if its joint density has a bivariate Gaussian form and the marginals are Gaussian.

---

b) Notation

$$p_{XY}(x,y) = P(X=x, Y=y)$$

$$p_X(x) = \sum_y p_{XY}(x,y), \quad p_Y(y) = \sum_x p_{XY}(x,y)$$

$$R_{XY} = E[XY]$$

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

$$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y]$$

$$\rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}}$$

Joint Gaussian PDF:

$$f_{XY}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\right.\right.\right.$$

Assumptions and Preliminaries

- X and Y are discrete or continuous random variables with valid joint distributions.
- Total probability sums to 1.
- Expectations $E[X], E[Y], E[XY]$ exist.
- Covariance uses centered variables (subtracting means).
- Correlation depends on scale and is computed about origin.
- Zero covariance implies uncorrelated but does not imply independence.

Main Result

a) Proof

Marginal PMF derivation:

$$p_X(x) = \sum_y p_{XY}(x,y), \quad p_Y(y) = \sum_x p_{XY}(x,y)$$

Covariance derivation:

$$\begin{aligned}\mathrm{Cov}(X,Y) &= E[(X-\mu_X)(Y-\mu_Y)] \\ &= E[XY - X\mu_Y - Y\mu_X + \mu_X\mu_Y] \\ &= E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X\mu_Y\end{aligned}$$

Since $\mu_X = E[X], \mu_Y = E[Y]$,

$$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y]$$

Properties:

1. Symmetry:

$$\mathrm{Cov}(X,Y) = \mathrm{Cov}(Y,X)$$

2. Variance:

$$\mathrm{Cov}(X,X) = \mathrm{Var}(X)$$

3. Linearity:

$$\mathrm{Cov}(aX+b, cY+d) = ac\,\mathrm{Cov}(X,Y)$$

4. Independence:

$$X \perp Y \Rightarrow \mathrm{Cov}(X,Y) = 0$$

Pearson Coefficient:

$$\rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X\sigma_Y}$$

$$-1 \le \rho_{XY} \le 1$$

---

b) Conclusion

- Marginal PMFs are obtained by summing over the other variable.
- Correlation $E[XY]$ mixes mean, spread, and dependence.

- Covariance measures centered dependence.
- Pearson coefficient normalizes covariance and gives strength and direction.
- Joint Gaussian variables follow a bivariate normal distribution with parameters $\mu_X, \mu_Y, \sigma_X, \sigma_Y, \rho$.

Working Examples

Example 1: Marginal PMF

Given:

$$p_{XY}(x,y): \qquad \begin{array}{c|cc} & y=0 & y=1 \\ x=0 & 1/8 & 1/4 \\ x=1 & 3/8 & 1/4 \end{array}$$

Steps:

$$p_X(0) = \frac{1}{8} + \frac{1}{4} = \frac{3}{8}, \quad p_X(1) = \frac{3}{8} + \frac{1}{4} = \frac{5}{8}$$
$$p_Y(0) = \frac{1}{8} + \frac{3}{8} = \frac{1}{2}, \quad p_Y(1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Final:

$$p_X = \{3/8, 5/8\}, \quad p_Y = \{1/2, 1/2\}$$

Example 2: Covariance

Given:

$$Y = -2X + 3$$

Using:

$$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y]$$
$$XY = X(-2X+3) = -2X^2 + 3X$$
$$E[Y] = -2E[X] + 3$$

Substituting:

$$\mathrm{Cov}(X,Y) = -2E[X^2] + 2(E[X])^2$$

Using:

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2$$

Final:

$$\mathrm{Cov}(X,Y) = -2\,\mathrm{Var}(X)$$

Example 3: Continuous Case

Given:

$$f_{XY}(x, y) = \frac{xy}{96}, \quad 0 < x < 4, 1 < y < 5$$

Results:

$$E[X] = \frac{8}{3}, \quad E[Y] = \frac{31}{9}$$

$$E[XY] = \frac{248}{27}$$

$$E[2X + 3Y] = \frac{47}{3}$$

$$\mathrm{Var}(X) = \frac{8}{9}, \quad \mathrm{Var}(Y) = \frac{92}{81}$$

$$\mathrm{Cov}(X, Y) = 0$$

$$\rho_{XY} = 0$$

Example 4: Joint Gaussian Interpretation

- X: CO_2 concentration
- Y: Temperature anomaly

Joint behavior approximated by Gaussian distribution due to multiple random factors.

If $\rho > 0$, then higher CO_2 corresponds to higher temperature.

END OF SCRIBE